GITAM Deemed to be University

Information Technology | Course Code: IT303

Cyber Security & Cryptography - Laboratory Manual & Practical Guide

Syllabus Focus: IT303 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Symmetric Cryptography & Block Ciphers

" Syllabus Scope: Feistel cipher model, DES, Triple-DES, Advanced Encryption Standard AES rounds, Modes o

Unit 2: Asymmetric Cryptography & Number Theory

" Syllabus Scope: Modular arithmetic, Extended Euclidean algorithm, RSA cryptosystem, Diffie-Hellman key

IT303: Cyber Security & Cryptography - Advanced Topics & Assessment

Academic Track: Information Technology - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Cryptographic Hash & Digital Signatures

" Hash functions, SHA-256, HMAC construction, Digital signatures DSA, ECDSA, Public Key Infrastructure PKI, X.

Unit 4: Network Security Protocols

" TLS 1.3 handshake, IPsec AH and ESP, Firewalls Stateful packet inspection, Intrusion Detection Systems IDS,

Unit 5: Web Security & Cyber Defenses

" OWASP Top 10 vulnerabilities, SQL injection, XSS, Buffer overflow exploits, Threat modeling, Zero-Trust netw

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Implementation of Classical Substitution and Permutation Ciphers in Python

Lab 2: Encryption and Decryption using AES-256-GCM with Authenticated Encryption Tags

Lab 3: Asymmetric Key Generation, Encryption, and Digital Signature Verification using RSA

Lab 4: Network Traffic Packet Inspection and Vulnerability Assessment using Wireshark and Nmap

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* William Stallings - Cryptography and Network Security: Principles and Practice Pearson 7th Ed

* Bruce Schneier - Applied Cryptography: Protocols, Algorithms, and Source Code in C Wiley

Authorized NoteVault Educational Document " Verified for Registered University Students